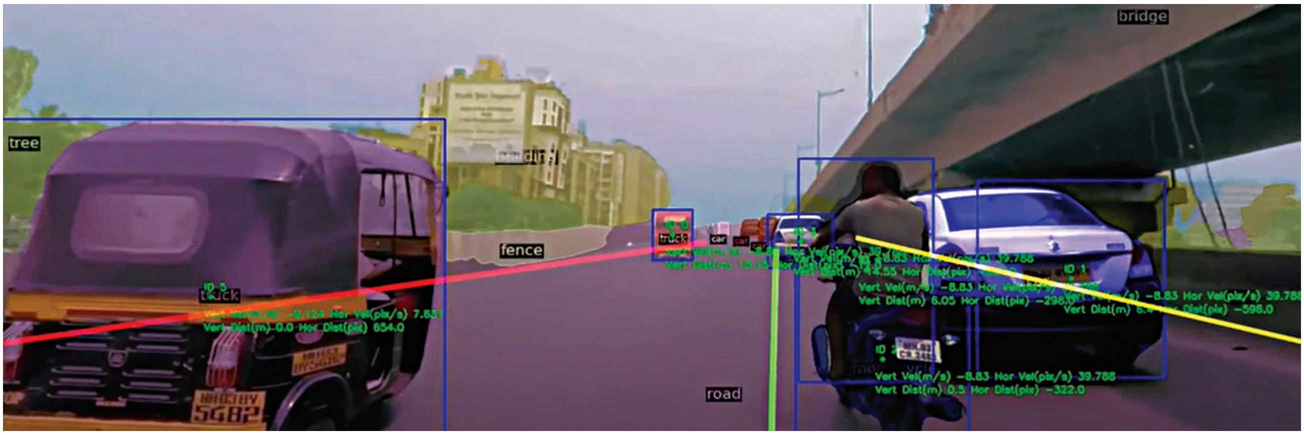

Minus Zero vehicle uses software to process information from its eight cameras to plan its own movement

technological hindrance associated with Indian traffic are there, that is where we have a USP because our tech is particularly suited for Indian traffic.

What is the difference between your tech and that of your counterparts?

Our biggest USP, in this case, is that our AI is less dependent on data. As a self-driving firm, you usually need more and more data for your system to increase its accuracy. So, data is both a boon and bane to the AI industry. All other self-driving companies require collection of a humongous amount of data for each city to make their system robust for that city.

This is where we have the advantage since we don't need as much data. Our system is a nature-inspired AI. Humans do not need to see innumerable images of a bird from various angles to understand what it is; one or two pictures are usually enough to render that information into our brain using which we can identify one when it is in front of us. Similarly, our system can extract 3D information from 2D data only. We have zero dependencies on lidar. What lidars do to generate 3D information, we can get the same 3D data from our camera feed only without compromising on accuracy.

In other words, it's primarily an edge based AI?

Yes, it is totally on edge. In fact, the e-rickshaw prototype that we shot the video of, the entire processing was done on a gaming laptop. It was a clear indicator that our system doesn't require huge processing power to make decisions. The target of the rickshaw prototype was three things. First was to show that it can handle Indian traffic. Second, to show that we do not need very high-end hardware to process that kind of data and, if you have an intuitive algorithm, the same thing can be worked on normal hardware. Third, this technology is not that expensive as the industry claims it to be.

You're still pursuing your engineering degree. So, where did you get the background to develop such a revolutionary tech?

While I am still studying engineering at the Thapar Institute of Engineering and Technology, I have been in the AI industry for a very long time. I started coding from grade 4 and by the time I graduated from high-school, I had already tested my hands on different types of technologies with AI in particular. You can often find me as a judge, mentor, or speaker on AI and autonomous robotics in various technical events, conferences, and colleges across works.

The idea of Minus Zero started as a research paper on 'AI that is less dependent on data,' but research limited to a pen and paper holds little value in real-life applications. We wanted to use this research for the benefit of the human race. That is when Gursimran (co-founder of Minus Zero) and I started exploring options where this technology would be most feasible, and we realised that self-driving cars are something that will have the most prominent benefits from this kind of AI.

Are your patents for software only or is there one for electronics hardware too?

There are for hardware configurations as well. People think that self-driving is only a software matter, but it is not. It is a software plus hardware problem. If you want to do Level 5 self-driving autonomy, then hardware also plays a major role.

How does your tech reduce dependence on data?

There are certain companies that have an end-to-end approach. An end-to-end approach is a very big neural network with billions of parameters. Where on one side you input all the sensor data, and on the other side you get the output. The bigger the network is, the more they will have to drive on the same lane over and over to capture the maximum number of possible scenarios. That is why, with an end-to-end approach, you need a lot of data to train on. You do not know what is going on behind neural networks and how the bits are being optimised. You just know that you need data to increase accuracy, so you keep on feeding more and more data. This is the end to end approach which is being followed by companies that are on Level 4 autonomy in certain countries.

Then comes the approach of companies like Tesla and ours. Instead of considering driving as one big task, we divided it into multiple subtasks. Instead of one task, we have multiple tasks, which include object detection, semantic segmentation, motion planning, and other small tasks. Our goal is that instead of perfecting that one big task, we try to perfect each small task individually, so that we know where exactly we need to improve, what exactly needs data, and what has already been perfected.

It is easier to perfect an object detection algorithm with less data than an entire driving task. In a driving task, many kinds of situations can happen, and every time the situation will be different. You will need more and more data for that but if, for instance, you have to perfect the object detection algorithm, that can be done with less amount of data. Tesla has also done the